Next-Day Reported Municipal Service-Demand Forecasting Across Four Cities: A Reproducible Benchmark with Simulated Allocation Evaluation

Nassar Naif Alsharif

Independent Researcher NassarAlsharif0@gmail.com

***Abstract*—We present a reproducible multi-city benchmark for next-day reported municipal service-demand forecasting and evaluate how forecast representations behave when embedded in a controlled request-equivalent allocation simulation. Short-horizon 311 forecasts are a natural planning input, but forecast accuracy is only a proxy for the decisions a forecast supports. The benchmark, the paper's primary artifact, is built from 36.8 million acquired service-request records (30.2 million retained after documented exclusions) across four heterogeneous U.S. municipal systems (New York, Chicago, San Francisco, Austin; 2020–2025), harmonized into common service families and joined with NOAA station weather; acquisition, exclusion, harmonization, and provenance are guarded and one-command reproducible. The allocation layer is a transparent simulation over abstract request-equivalent capacity units: it concerns reported demand, not social need, and makes no claim about any city's real capacity, productivity, or operations. Budgets are frozen from training data and all model selection uses validation data only. Calendar augmentation reduces next-day error in every city (10.0–14.0%, all 28-day block-bootstrap intervals excluding zero); target-day weather — a day-ahead-forecast proxy — helps three cities and adds nothing in the fourth, a retained null; temporally censored cross-city pooling raises error in three of four cities, robust to log-scale and per-city-standardized normalization. Coupling forecasts to the simulation yields the paper's headline finding, a forecast-to-decision phenomenon: a degenerate point forecast leaves the equal request-equivalent objective non-identified across nearly the entire feasible set (92–100% of allocation steps are ties), so the realized outcome is set by the secondary tie-break rule. Under the naive fixed-index default this loses to proportional apportionment in all 12 city-regime settings; a proportional secondary rule recovers proportional performance, and a genuine quantile-interpolated predictive distribution removes most of the non-identification gap. When forecasts feed a discrete coverage-type objective with symmetric per-unit value, predictive representation and tie-breaking become first-class design choices rather than implementation details — a bounded, simulation- internal lesson, not a deployment or fairness claim. Aggregate simulated- loss reductions are not fairness guarantees: some low-volume families receive near-zero served fractions under this objective. All data are public and every number regenerates from one command under a guard suite that enforces the protocol.**

I. Introduction

Municipal 311 systems receive millions of service requests each year — heating failures, water leaks, street damage, sanitation problems, noise — and agencies plan finite daily service capacity against this demand. Short-horizon forecasts of request volume are a natural planning input, and a growing literature shows that public signals such as weather and calendar structure improve such forecasts. But planners do not consume mean absolute error; they consume allocations, and the mapping from predictive accuracy to allocation quality is neither linear nor guaranteed. We present a reproducible multi-city benchmark for next-day reported municipal service-demand forecasting and evaluate how forecast representations behave when embedded in a controlled request-equivalent allocation simulation.

The contributions form an explicit hierarchy rather than co-equal claims. The **primary contribution is the artifact**: a reproducible four-city benchmark of next-day reported 311 demand (New York, Chicago, San Francisco, Austin; 2020–2025) on a common service-family schema, built from official open-data APIs with versioned harmonization, provenance manifests, SHA-256 checksums, and leakage-controlled temporal evaluation, regenerable by one command. The artifact is a benchmark (data, protocol, and automated enforcement), not a data collection alone; its evaluation protocol includes the controlled request-equivalent allocation simulation as a measurement instrument of the benchmark, not a standalone simulation study. The **headline empirical finding** is what that instrument reveals: a degenerate point forecast leaves the equal request-equivalent allocation objective non-identified (92–100% of allocation steps tied), so the realized outcome is set by the secondary tie-break rule; a proportional secondary rule recovers proportional performance and a genuine predictive distribution removes most of the non-identification gap. **Supporting evidence** characterizes the forecasting layer: leakage-controlled calendar, weather, and public-signal value; local versus pooled and temporally censored cross-city transfer; uncertainty representation; and a count-data (Poisson) baseline, with negative and null findings reported transparently. Responsible use is enforced as a property of the protocol rather than claimed as a contribution: reported demand is never interpreted as social need, aggregate simulated loss is never presented as a fairness guarantee (family-level served fractions are disclosed), and every number is provenance-tracked and guard-enforced.

The single research question of the study, the question this instrument is built to answer, is:

> Across heterogeneous municipal 311 reporting systems, under pre-specified hypothetical service-pressure regimes, when do differences in next-day forecasts of reported administrative demand produce materially different outcomes in a transparent simu-

lated family-level capacity allocation, and when are predictive rankings sufficient proxies for simulated decision rankings?

Four design commitments distinguish the protocol (Section 5): frozen training-only capacity budgets, identical at selection and test; validation-only model selection; temporal censoring so pooled and transfer models never train on rows contemporaneous with any evaluation window; and a controlled uncertainty contrast that varies only how one fitted quantile model's median, implied-mean, and full-distribution representations enter the policy. A guard suite of automated tests enforces each commitment and fails the build on violation.

The contribution is one of data and evaluation, not a new algorithm, and it sits in the big-data tradition: *Volume* — tens of millions of acquired municipal records (36.8M acquired, 30.2M retained), entering through acquisition, exclusion, harmonization, and provenance, while the predictive task is defined on daily aggregated city–family panels, not raw records; *Variety* — four heterogeneous 311 systems with distinct taxonomies, harmonized families, calendar, and NOAA weather; *Veracity* — provenance manifests, documented exclusions, SHA-256 checksums, leakage guards, and one-command reproduction; *Value* — whether forecasting gains do or do not translate into simulated allocation outcomes. Velocity is out of scope: the modeling unit is the daily aggregate and we claim no real-time, streaming, or deployed system.

Throughout, we are explicit about what the data can and cannot support. Reported requests are not social need [1]; the allocation layer is a simulation over abstract request-equivalent capacity units (a simulation unit, not a real workforce measure), with parameters varied over a pre-specified grid rather than asserted; and no causal, deployment, or operational-guidance claim is made. The study bridges the forecasting literature, which stops at accuracy, and decision-focused learning [2, 3].

The paper proceeds from positioning (Section 2) through formulation, data, and methods (Sections 3–5) to forecasting results (Section 6), the simulated allocation evaluation (Section 7), robustness (Section 8), and responsible use, limitations, and conclusions (Sections 9–11).

II. Related Work

a) Public signals in operational forecasting.: Cui et al. [4] established, in retail, that publicly available signals (social media) measurably improve daily operational forecasts; Steinker et al. [5] did the same for weather in e-commerce fulfilment. The present work brings this template to municipal reported demand at multi-city scale and extends it in kind: four heterogeneous 311 systems under one leakage-controlled protocol; local-versus-pooled and temporally censored cross-city transfer; forecasting coupled to a controlled request-equivalent allocation simulation rather than stopping at accuracy; characterization of point-forecast non-identification, its tie-break dependence, and the family-level served-fraction tradeoffs it exposes; and a provenance- and guard-backed reproducible benchmark.

b) 311 analytics.: 311 data underpin a largely single-city, prediction-centric literature: Xu et al. [6] forecast Chicago 311 demand in space and time with an adaptive kernel and motivate the exercise by resource-allocation relevance without evaluating an allocation; Cheng et al. [7] predict request volumes and their correlates in Miami-Dade County; Kontokosta et al. [8] and Kontokosta and Hong [1] document socio-spatial disparities in the propensity to report, establishing that 311 volumes measure *reported* demand rather than need. We inherit the latter caveat explicitly. To our knowledge no prior work evaluates 311 forecasting across multiple cities under one protocol, nor couples it to an explicit allocation loss with frozen pre-test environments.

c) From prediction to decisions.: That predictive accuracy and decision quality are distinct objectives is the founding observation of prescriptive analytics [9] and decision-focused learning: the SPO framework trains models against the downstream objective [2], task-based end-to-end learning does so in stochastic programs [10], and Mandi et al. [3] survey and benchmark the field. Those benchmarks are dominated by synthetic costs or energy/inventory/routing settings; the same holds for the standard predict-then-optimize tooling, whose suites are shortest-path, knapsack, and travelling-salesperson instances with generated costs [11], and the contextual-optimization survey of Sadana et al. [12] likewise notes the scarcity of realistic public testbeds. We contribute complementary public-sector evidence: rather than proposing a training method, we measure empirically where accuracy-optimized models, distributional information, and decision-aware validation selection do and do not change simulated allocation outcomes.

d) Capacity planning under time-varying demand.: Setting service capacity against predictable daily variation is a classical operations problem [13], and recent public-sector work couples coherent point and probabilistic demand forecasts to deployment planning in emergency medical services [14]. Our simulation is a deliberately transparent discrete-time instance with integer capacity units, family-level objectives, and carryover of unresolved request-equivalents, designed for auditability rather than queueing fidelity.

e) Cross-city learning and urban computing.: Urban computing treats the city as a sensing-and-decision platform [15]; cross-city transfer has been studied mainly for traffic and crowd-flow prediction [16], and pooled training across related series is standard in modern forecasting [17]. Recent urban benchmarks operate at this platform scale: Liu et al. [18] release a five-year, 8,600-sensor traffic-forecasting benchmark, and Ning et al. [19] build two-city (New York, Chicago) urban knowledge-graph datasets whose downstream tasks include 311 request prediction; both evaluate predictive accuracy only. We test pooled-versus-local training and zero-shot transfer under explicit temporal censoring, and carry the comparison through to simulated decision quality. Table I summarizes the positioning.

TABLE I
POSITIONING AGAINST REPRESENTATIVE PRIOR WORK.

Study	Data scope	Evaluation focus	Forecast→allocation layer
Xu et al. [6]	Chicago 311 (one city)	space–time demand prediction	motivated, not evaluated
Cheng et al. [7]	Miami-Dade 311 (one county)	request volumes and correlates	none evaluated
Ning et al. [19]	NYC+Chicago knowledge graphs	spatiotemporal prediction incl. 311	none evaluated
Liu et al. [18]	California traffic (8,600 sensors)	large-scale traffic forecasting	none evaluated
Mandi et al. [3]	stylized/synthetic decision tasks	decision-focused training methods	in the training objective
This work	four 311 systems, 36.8M records	leakage-controlled forecasting protocol	controlled allocation simulation, tie-break analysis

III. PROBLEM FORMULATION

a) Data in brief.: The unit of analysis is the calendar day. For each of the four cities the benchmark provides a daily panel: the count of 311 requests created in city c and harmonized service family s on day t, joined with deterministic calendar information and one station's weather observations. Section 4 details sources, exclusions, harmonization, and provenance; everything below is defined on this panel.

b) Prediction task.: Input: the panel through day t. Output: next-day forecasts for every active family. Goal: measure what each public signal adds, under strict leakage control. Let $y_{c,s,t}$ denote the number of 311 requests created in city c and harmonized service family $s \in \mathcal{S}_c$ on calendar day t, where $\mathcal{S}_c$ is the city's *active* family set (eight families in New York, San Francisco, and Austin; seven in Chicago, where residential noise is structurally absent from the intake taxonomy). At the end of day t (the information cutoff), a forecaster produces a point forecast $\hat{y}_{c,s,t+1}$ and, for the probabilistic model, quantile forecasts $\hat{q}^{\tau}_{c,s,t+1}$, $\tau \in \{.05, .25, .50, .75, .95\}$. Endogenous features use observations through day t only; calendar features of day $t{+}1$ are deterministic; target-day weather uses the realized observation for $t{+}1$ as a stand-in for a day-ahead weather forecast (assumption A3, bounded by a lagged-weather variant).

c) Why a simulated allocation layer.: Accuracy metrics alone cannot reveal whether differences between forecasts would ever change a decision. The benchmark therefore couples every forecaster to one deliberately transparent allocation simulation: an instrument for measuring the forecast-to-decision mapping under a known objective, every parameter exposed as configuration, every step auditable. It models no city's actual operations; any difference in simulated outcome is attributable, by construction, to the forecasts and the policy rule.

d) Simulated allocation.: Input: the day-$t{+}1$ forecasts and the observable carryover state. Output: an integer split of a fixed capacity budget across families. Goal: score each forecast by the unresolved workload its allocations leave behind. Each evening a simulated planner allocates a fixed budget of B_c abstract capacity units across active families, $x_{c,s,t+1} \in \mathbb{Z}_{\geq 0}$, $\sum_s x_{c,s,t+1} = B_c$; a unit resolves up to κ request-equivalents the next day. Unresolved workload carries over:

$$\begin{aligned} w_{c,s,t+1} &= b_{c,s,t} + y_{c,s,t+1}, \\ \text{resolved} &= \min(w_{c,s,t+1},\, \kappa\, x_{c,s,t+1}), \\ u_{c,s,t+1} &= w_{c,s,t+1} - \text{resolved}, \\ b_{c,s,t+1} &= (1-\alpha)\, u_{c,s,t+1}, \end{aligned} \tag{1}$$

where b is simulated unresolved request-equivalent carryover (observable state at decision time), initialized at $b_{c,s,0} = 0$, and α an abandonment rate ($\alpha = 0$ primary; b_0 and α are varied as labeled sensitivities). In words: each day's workload w is new requests plus carried stock; allocated capacity resolves up to κ request-equivalents per unit; whatever remains unresolved (u) carries into the next day's state b. The decision loss over the test horizon $\mathcal{T}$ is

$$L_c \;=\; \sum_{t \in \mathcal{T}} \sum_{s \in \mathcal{S}_c} \omega_s\, u_{c,s,t}, \tag{2}$$

with EQUAL request-equivalent weights $\omega_s = 1$ as the primary objective — a deliberately simple equal-weight choice, not a normatively neutral or normative priority policy (one illustrative normative-priority scenario is a labeled sensitivity). Because $u_{c,s,t}$ contains the carried stock, a request that remains unresolved for k days contributes k times to (2): the loss is a *waiting-time-weighted cumulative unresolved stock*, not a daily flow. Under the scarce regime the simulated system is structurally under-capacitated (supercritical), so this stock grows over the horizon and daily losses — and the daily loss *differentials* between policies — can carry a deterministic trend; Section 8 reports a trend diagnostic and the inference framing follows it. Hypothetical service-pressure regimes set κB_c to 70%, 90%, and 110% of the city's TRAINING-window mean daily demand $\bar{d}_c$: that is, $B_c = \max\big(\text{round}(f\,\bar{d}_c/\kappa),\, |\mathcal{S}_c|\big)$ for $f \in \{0.7, 0.9, 1.1\}$, floored at the active-family count. The resulting integer budgets are frozen before any model selection and used unchanged for validation-stage selection and the single test evaluation; κ sets only the scale of a capacity unit. Rather than sweeping a global κ grid, the sensitivity analysis varies one family's service-yield multiplier at a time (factors 0.70 and 1.30). Nothing in this construction estimates any city's actual capacity, productivity, or queues.

e) Policies.: Input: the same observable state for every policy. Output: a full allocation of the budget. Goal: isolate how the forecast's representation, not its information set, changes the allocation. All policies receive the same observable state and allocate the full budget: *uniform* (equal units; forecast-free floor); *proportional* (largest-remainder apportionment of

$\hat{y}_{c,s,t+1} + b_{c,s,t}$); and *greedy expected-value*, which exactly maximizes $\sum_s \omega_s \mathbb{E}[\min(D_s, \kappa x_s)]$ over integer allocations, where D_s is the forecast demand-plus-carryover distribution — degenerate for point forecasts, piecewise-linear from the quantile model. Greedy is optimal for this objective because $\mathbb{E}[\min(D, c)]$ is concave in c; ties in marginal value are resolved by a fixed secondary key (ascending family index). Under a degenerate point forecast this objective is *non-identified* across most of the feasible set, so the realized allocation — and hence its loss — depends on that key; we treat the key as a first-class object and report its effect in Section 7. A hindsight myopic reference (greedy on realized demand) appears in supplementary diagnostics only; being per-day myopic it is not a horizon-optimal bound and is never used as a denominator or target.

f) Controlled uncertainty contrast.: Input: one fitted quantile model. Output: three policy arms that differ only in the representation handed to the same policy. Goal: isolate the value of distributional information. The value of distributional information is isolated within ONE fitted quantile model: a degenerate-median arm, an implied-mean arm, and a full-distribution arm differ only in how the same predictive distribution enters the greedy policy. The predictive distribution is reconstructed by linear interpolation of the inverse CDF through the five forecast quantiles on a fixed 99-point grid on $[0.01, 0.99]$, with flat (clamped) tails beyond the .05 and .95 levels; the implied mean is the average of the 99 grid samples and the median arm is the .50 quantile. We write "quantile-interpolated predictive distribution" for this object throughout. The separately trained point models serve as forecasting benchmarks, not as the uncertainty comparator.

g) Evaluation questions.: For each city and regime we compare forecast accuracy (MAE, with paired 28-day moving-block bootstrap intervals; pinball loss and coverage for quantiles) and simulated decision loss (2) (raw losses, absolute paired daily differences with block-bootstrap intervals, and percentage reduction relative to the uniform floor), asking when predictive and decision orderings agree, whether distributional information changes outcomes, and whether validation-time selection by decision loss differs from selection by accuracy.

IV. Data

a) 311 service requests.: Input: official open-data portals. Output: versioned daily aggregates with full provenance. Goal: an acquisition layer any reader can re-run and audit. We use the official open-data portals of four U.S. cities, selected by pre-stated criteria (API uniformity, coverage of the 2020–2025 study window, category granularity, and heterogeneity in size and climate; see the feasibility audit in the companion repository): New York (`erm2-nwe9`, NYC Open Data), Chicago (`v6vf-nfxy`, current 311 system, live since December 2018), San Francisco (`vw6y-z8j6`, DataSF, PDDL license), and Austin (`xwdj-i9he`). Acquisition uses server-side daily aggregation (counts per day × native category) through the SODA API; every query URL, retrieval timestamp, and SHA-256 checksum is recorded in versioned manifests. Chicago rows flagged as duplicates by the source system are removed (523,055 requests), as are documented non-service categories: Chicago's "311 INFORMATION ONLY CALL" log entries (4,184,157 records) and aviation-routed aircraft-noise filings (1,918,574 records), and San Francisco's administrative duplicate markers. The 311 acquisition and study window is 2020-01-01 through 2025-12-31, the full scope of NYC's current dataset, and the densified panel spans this window. The *modeled span* is shorter and is fixed by a frozen weather-joined evaluation panel: every feature set shares one panel that includes station weather and rows lacking weather are dropped, so the modeled span — and hence all forecasting and decision evaluation — runs 2020-01-28 through 2025-02-05. The NOAA GHCN-Daily layer acquired for this benchmark covers the four stations through 2025-02-06, whereas the 311 layer covers through 2025-12-31; the later 311 observations are therefore retained in the panel but lie outside this frozen common evaluation window and are not modeled. As an acquisition-integrity check, the NYC layer was produced by two independent methods — server-side aggregation and client-side aggregation of a streamed export — which agreed exactly (270,978 aggregated rows).

b) Harmonized service families.: Input: four native category taxonomies. Output: one eight-family schema. Goal: cross-city comparability without claiming compositional identity. Each city's native taxonomy (∼one to three hundred categories) is mapped to eight service families — noise; housing/building; water/sewer; public safety; streets/transport; sanitation; parks/trees; other — by an ordered, case-insensitive, versioned keyword rule set applied identically to all cities. The rules were audited against each city's real category inventory and frozen before any model was trained; after the freeze, the unmapped "other" share is 6.2% (New York), 6.9% (Chicago), 8.8% (San Francisco), and 23.5% (Austin, whose intake includes large animal-services and explicitly generic categories that fit no family and are deliberately retained in the served "other" bucket). Families are *operationally analogous planning buckets*, not identical workloads: Chicago has no noise family at all (residential noise is not a Chicago 311 request type), the supplement reports the full native-category composition of every family in every city, and no cross-city claim in this paper assumes compositional identity.

c) Weather.: Input: NOAA GHCN-Daily station records. Output: daily weather features per city. Goal: a bounded, checksummed public weather signal. NOAA GHCN-Daily station observations (precipitation, max/min temperature, snowfall, snow depth; derived mean temperature) from one first-order station per city (Central Park; O'Hare; San Francisco Downtown; Camp Mabry), retrieved from NOAA's AWS Open Data mirror with checksums. Quality-flagged values are dropped; short temperature gaps (≤ 7 days) are interpolated; missing snow fields are treated as zero. Wind speed was excluded for all cities on data-completeness grounds: the San Francisco station does not report it and Central Park has a multi-month gap; the exclusion is recorded in the repository's decision log. A single station per city is a documented spatial

TABLE II

DATASET SUMMARY AFTER PREPROCESSING (STUDY WINDOW 2020-01-01 TO 2025-12-31; EIGHT DEFINED SERVICE-FAMILY CATEGORIES WITH CITY-SPECIFIC ACTIVE SETS — CHICAGO HAS SEVEN BECAUSE THE NOISE FAMILY IS STRUCTURALLY ABSENT; NO SYNTHETIC DATA). COUNTS, PANEL ROWS, AND PER-DAY MEANS ARE FOR THE DENSIFIED STUDY WINDOW OVER THE FULL ACQUIRED/RETAINED 2020–2025 REQUESTS; FORECASTING AND DECISION EVALUATION USE THE SHORTER MODELED SPAN — THE FROZEN WEATHER-JOINED COMMON EVALUATION PANEL — THROUGH 2025-02-05, SO NOT ALL RETAINED REQUESTS ENTER MODELING (SEE TEXT).

City	Requests kept	Native cat.	Panel rows	Days	Mean/day
Austin	1,487,588	346	17,536	2192	679
Chicago	4,683,510	107	15,344	2192	2,137
New York	19,669,371	271	17,536	2192	8,973
San Francisco	4,374,676	136	17,536	2192	1,996

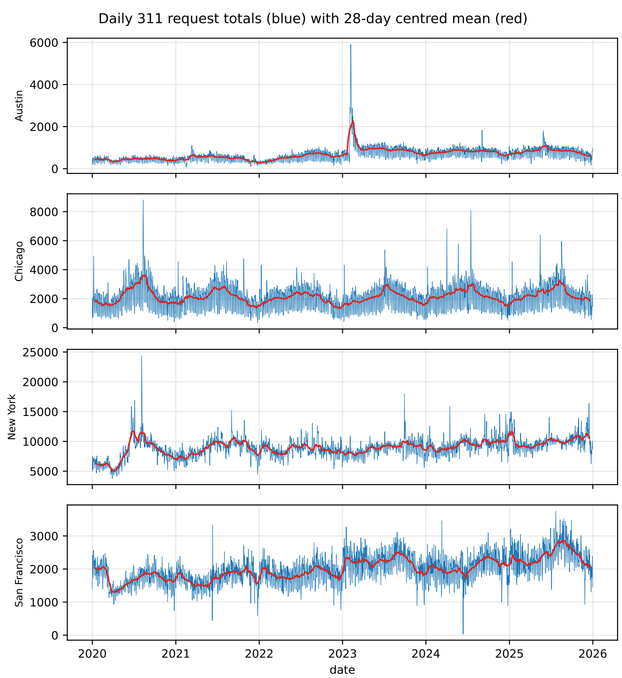

Fig. 1. Daily citywide 311 request totals (blue) with 28-day centred mean (red), 2020–2025. Scale, seasonality, and disruption patterns differ visibly across the four systems.

simplification (A2). No synthetic demand or synthetic weather is used anywhere in the study.

V. METHODS

a) Feature sets.: Input: the daily panel. Output: four nested predictor sets. Goal: isolate the marginal value of each public signal. Four nested feature sets isolate the marginal value of each public signal: *internal* (lags $1, 2, 3, 7, 14, 28$ and 7/28-day rolling mean and standard deviation, windows ending at day t); *calendar* (internal + deterministic day-of-week, month, weekend, U.S. federal holiday, and smooth day-of-year terms for the target day); *weather* (internal + target-day and current-day station weather); and *calendar+weather*. Family indicators are one-hot encoded; pooled models add city indicators.

b) Models.: Input: the feature sets. Output: a pre-specified grid of point and quantile forecasters. Goal: reproducible comparison, not a leaderboard. Two forecast-free baselines (trailing 7-day mean; same-weekday seasonal naive), ridge regression, a random forest [20], and LightGBM with an L1 objective [21] form the pre-specified point-model grid; quantile-objective LightGBM [following 22] provides the predictive distribution, with non-crossing enforced by cumulative maxima. Hyperparameters are fixed across cities and feature sets; baselines are never dropped from tables.

c) Protocol.: Input: each city's day sequence. Output: splits, model selections, and one test evaluation per configuration. Goal: leakage-controlled selection that never touches test outcomes. Within each city, the earliest 70% of days form training, the next 15% validation, and the final 15% an untouched test window. For every (city, feature set), the model is selected on VALIDATION MAE, frozen, refit on train+validation, and evaluated once on test; exhaustive test grids appear only as supplementary descriptive evidence. The quantile model is fit on training data alone for all validation-stage outputs and refit on train+validation for its single test evaluation. Stability is assessed with five expanding-window rolling-origin folds [23]. Pooled ("global") models train on all four cities with stage-specific temporal censoring: validation-stage fits use only rows dated no later than the earliest training cutoff across cities, test-stage fits no later than the earliest validation cutoff, so no pooled training row is contemporaneous with any evaluation window. Leave-one-city-out transfer is likewise censored at the held-out city's validation cutoff. Two selection procedures with explicitly different candidate sets are used and named throughout: *within-scope selection* (per city, feature set, and scope, used for every headline table and figure) selects among the five point models of that scope only; the *cross-scope selection experiment* (Section 7) selects over a pre-specified eleven-configuration grid that pools the ten local configurations with the stage-censored global calendar+weather model. All accuracy contrasts use paired moving-block bootstrap intervals on daily mean absolute-error differentials (28-day blocks, 2,000 resamples, seed 20260609; 14- and 56-day blocks as pre-specified sensitivity) [in the spirit of 24].

d) Decision evaluation.: Input: every forecaster's next-day forecasts. Output: simulated decision losses at frozen budgets. Goal: compare policies and representations on the decision loss, not only on accuracy. Every point forecaster drives the proportional and greedy policies; the quantile model drives the three uncertainty arms; uniform allocation is the forecast-free floor. Simulations run day-by-day over each city's test window at the frozen budgets with $\kappa = 50$ and equal weights. Pre-specified sensitivities: one normative-priority scenario; one-family-at-a-time service-yield multipliers of 0.70 and 1.30 over every active family; 10% abandonment; train+validation budget calibration; and an Austin variant that excludes the heterogeneous *other* family with budgets recomputed accordingly. Validation-time model selection by decision loss (simulated on the validation window at the same frozen budgets) is compared against selection by validation MAE, both frozen before test.

e) Guard suite.: Input: the repository's data, configuration, and outputs. Output: 37 automated checks. Goal: fail the build on any protocol violation. Sixteen automated protocol guards (G1–G16), exercised by a 37-test suite, enforce the protocol and fail the build otherwise, including: budgets reproducible from training data and invariant to test perturbation; selection invariant to test metrics; recomputed censoring dates on every transfer row and a fit-time proof manifest for pooled censoring; identical fitted model behind all uncertainty arms; structural absence never encoded as zero; exact budget conservation; forbidden-terminology scans of all manuscript-facing artifacts; configuration–output agreement for every sensitivity scenario; source-identifier consistency; and provenance hashes binding every table and figure to the generating run.

VI. Forecasting Results

All headline aggregations use validation-only selection; exhaustive grids are supplementary. Intervals are paired 28-day moving-block bootstrap (stable at 14 and 56 days throughout).

a) Calendar gains replicate everywhere.: The validation-selected calendar model lowers held-out test MAE relative to the validation-selected internal-history model in every city: -11.9% in New York (207.0 $\rightarrow$ 182.4), -15.6% in Chicago (60.4 $\rightarrow$ 50.9), -10.4% in San Francisco (35.2 $\rightarrow$ 31.5), and -10.2% in Austin (16.5 $\rightarrow$ 14.8) — a validation-selected comparison spanning 10.2–15.6%. As a separate comparison holding the estimator fixed, the LightGBM calendar effect (internal $\rightarrow$ calendar, same model) is 10.0–14.0% and consistent in all four cities, all 28-day block-bootstrap intervals excluding zero. Against the same-weekday seasonal-naive baseline, the selected calendar+weather model reduces test MAE by 41.4% (New York), 41.5% (Chicago), 31.3% (San Francisco), and 31.3% (Austin). Validation-only selection matters in its own right: on the internal feature set it picks the random forest in Austin and Chicago and ridge in San Francisco, choices a test-minimum rule would have hidden. As a count-data check, a Poisson GLM (supplement) beats the naive trailing mean in 11 of 12 (city, feature-set) cells but is dominated by the validation-selected models in all 12; consistent with this being a benchmark-and-protocol contribution rather than an algorithmic leaderboard, we add no further models here and defer modern deep or hierarchical baselines to a journal version.

b) Weather's marginal value is real but city-dependent.: On top of calendar features, target-day weather further reduces MAE by 9.3% in New York, 3.3% in Chicago, and 4.7% in San Francisco (all intervals excluding zero) — and by exactly nothing in Austin ($+0.001$ MAE; interval covers zero). The Austin null is retained as a finding. The lagged-weather variant (assumption A3 removed) erases most of New York's weather gain (178.6 versus 165.4, against a 182.4 calendar-only reference), so much of the weather value is conditional on an accurate day-ahead weather forecast — a realistic but explicitly stated premise.

c) Stability.: Five expanding-window rolling-origin folds reproduce the feature-set ordering (internal $>$ calendar $>$ calendar+weather in MAE) in every city; the headline comparisons are not artifacts of a single split.

d) Pooling and transfer mostly hurt under valid censoring.: With stage-censored training (no pooled training row contemporaneous with any evaluation window), the pooled model raises test MAE relative to local models in Austin ($+4.1\%$), Chicago ($+6.5\%$), and New York ($+1.8\%$), each interval excluding zero, and is neutral in San Francisco (-1.2%, interval covers zero). Two distinct selection procedures operate in this paper and must not be conflated: the *within-scope* headline selection (Section 5) chooses among local candidates for the local headlines, where the local LightGBM wins in every city; the *cross-scope* selection experiment of Section 7 chooses over the full eleven-configuration grid, where validation MAE still prefers the pooled model in Austin (validation MAE 15.08 vs. 15.25 local, despite worse test error) and in San Francisco (27.20 vs. 28.97, where the pooled model is also genuinely better on test). Temporally censored zero-shot transfer (training without the target city, censored at its validation cutoff) degrades MAE by $+5.4\%$ (San Francisco) to $+148.5\%$ (New York). The pooling result is not an artifact of pooling raw counts across cities of very different scale: refitting the pooled model on a $\log(1+y)$ target or on per-city-standardized targets shrinks the penalty (most for the smallest city, Austin: $+4.1\%$ $\rightarrow$ $+2.0\%$) but does not reverse it — pooling still raises test MAE in three of four cities under both normalizations, with San Francisco the lone neutral case. With six years of local history, cross-city pooling and transfer are not beneficial in this benchmark; the negative result, robust to target normalization, is retained.

e) Probabilistic forecasts are sharp but under-dispersed; conformal recalibration restores coverage.: The quantile model's central 90% intervals cover 76.8–81.2% of test outcomes. Split-conformal recalibration using validation residuals only (per city, per quantile level) raises empirical 90% coverage to 88.3–90.3% while widening intervals modestly. We report both the raw miscalibration and its conformal correction; Section 7 shows that the distributional information identifies the simulated allocation objective whether or not the intervals are recalibrated (the calibrated full arm changes simulated loss by a median of -0.1%), so the decision result reflects identification rather than the particular dispersion of the intervals.

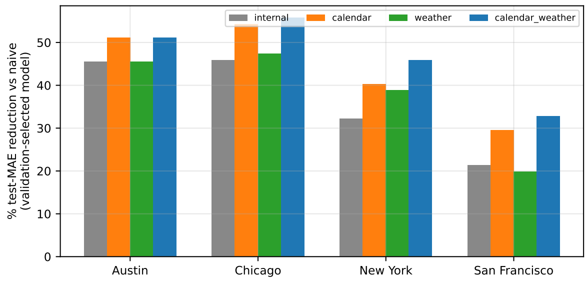

Fig. 2. Held-out test-MAE reduction versus the trailing-mean baseline for the validation-selected model of each feature set and city.

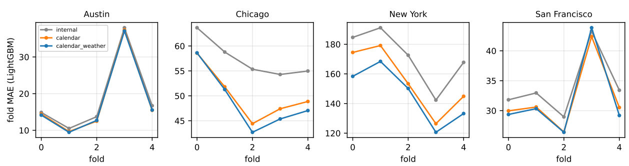

Fig. 3. Rolling-origin fold MAE (LightGBM) by feature set: the calendar and weather orderings are stable across folds in all cities.

VII. Simulated Capacity-Allocation Evaluation

All simulations run at budgets frozen from training data (New York 119/153/187 units across the scarce/moderate/generous regimes; Chicago 29/38/46; San Francisco 26/33/41; Austin 8/11/13; $\kappa = 50$), under the equal request-equivalent-weights objective. Losses are waiting-time-weighted unresolved stocks (Section 3); policy contrasts carry paired 28-day block-bootstrap intervals on the daily loss differentials. Because the trend diagnostic of Section 8 finds 45 of the 48 differential series trend-dominated (the carryover stock compounds policy differences), we read these intervals as *descriptive characterizations of a systematic, partly deterministic divergence within the simulation*, retaining a stationary-inference reading only where the diagnostic supports it (Chicago, generous regime). The simulation itself is deterministic given the frozen inputs, so every directional statement below is exact for this environment; what the trending differentials preclude is generalizing the interval widths as stationary sampling uncertainty. Three findings emerge.

a) Degenerate point forecasts non-identify the allocation objective; the realized outcome is tie-break dependent.: Degenerate point-forecast representations collapse the marginal-value profile of the myopic expected-value allocation objective. Under the equal-weight objective the per-unit marginal value equals $\kappa\omega$ for every unit below the point estimate plus carryover and zero above it, so all under-served families share an identical marginal value and the greedy ordering carries no information in either the deficit or the surplus segment (worked examples for both segments are in the supplement). That ties arise under a symmetric coverage objective is not in itself surprising; the empirical question is whether this non-identification is rare enough to ignore or pervasive enough to govern the downstream conclusion. On real multi-city municipal data it is pervasive: the objective is non-identified across nearly the entire feasible set — 92–100% of allocation steps are ties, with 2.6–6.9 active families tied at the maximum marginal value (Table III), across four heterogeneous demand structures. Realized allocation loss is therefore determined by the secondary tie-break rule, not by the point forecast. Under our original implementation the fixed ascending-index tie-break underperformed proportional apportionment in all 12 city-regime cells (by 1.2–58%); a random tie-break is intermediate and a Gaussian/Poisson smoothing of the point forecast only partly recovers performance. A *proportional* secondary tie-break — a representation-free regularization toward proportional apportionment — fully recovers proportional performance (-0.03% to $+0.8\%$ across the 12 cells). The issue is thus representation-induced *non-identification* of the allocation objective under degenerate forecasts, not mathematical infeasibility of the optimizer. Relative to the forecast-free uniform floor, the proportional policy with the selected forecaster reduces the cumulative unresolved stock by 18–23% (scarce), 38–70% (moderate), and 60–93% (generous), the regime pattern itself being partly mechanical: under structural under-capacity no policy can prevent stock growth, so policy differences compress. These aggregate simulated-loss reductions are not fairness guarantees: the responsible-use discussion and the supplement show that some low-volume families receive near-zero served fractions under the equal request-equivalent objective. All of this is a property of the simulated environment and its equal-weight objective — under the normative-priority sensitivity the gap compresses — and constitutes no guidance about any city's actual operations. The design lesson is bounded but real: when forecasts feed a discrete coverage-type objective with symmetric per-unit value, predictive representation and the secondary tie-break rule are first-class design choices, not implementation details. We state this only for the tested objective, regimes, reported municipal demand, and simulated request-equivalent allocation; it is a forecast-to-decision design phenomenon, not deployment evidence.

b) A genuine predictive distribution removes the non-identification, a second route alongside a proportional tie-break.: Holding the single fitted quantile model and every other input fixed, the quantile-interpolated full-distribution arm outperforms both the median arm and the implied-mean arm in 12 of 12 settings *under the fixed-index tie-break*; the paired blocked-resampling summaries are reported descriptively and, in this comparison, do not cross zero. This advantage is the same non-identification phenomenon: the median and implied-mean arms are themselves degenerate point forecasts and inherit the same flat-profile ties, whereas the full distribution's tails supply a genuine, non-flat marginal-value profile. When the median and implied-mean arms are instead resolved with the proportional secondary tie-break, the full-distribution advantage disappears entirely (full better in 0 of 12 cells, by -0.6% at the median); both then match proportional. The defensible claim is therefore narrow and is the paper's headline finding, the result the benchmark's decision-evaluation instrument exists to measure: forecast *representation* matters because degenerate

TABLE III

OBJECTIVE NON-IDENTIFICATION UNDER DEGENERATE POINT FORECASTS. "TIE %" IS THE SHARE OF ALLOCATION STEPS WITH MORE THAN ONE ACTIVE FAMILY TIED AT THE MAXIMUM MARGINAL VALUE. THE REMAINING COLUMNS GIVE THE SIMULATED LOSS OF THE POINT-FORECAST GREEDY POLICY UNDER FIVE TIE-BREAK RULES, AS % RELATIVE TO PROPORTIONAL APPORTIONMENT (NEGATIVE = WORSE THAN PROPORTIONAL). THE IDENTICAL POINT FORECAST RANGES FROM FAR WORSE (FIXED ASCENDING-INDEX) TO INDISTINGUISHABLE FROM PROPORTIONAL (PROPORTIONAL SECONDARY RULE); THE FULL QUANTILE-INTERPOLATED DISTRIBUTION RECOVERS MOST OF THE GAP. FULL GRID IN THE SUPPLEMENT.

City	Regime	Tie %	Fixed	Rand. worst	Rand. best	Prop. TB	Full dist.
austin	generous	99.80	-3.61	-0.93	-0.53	0.19	-0.27
austin	moderate	99.90	-2.04	-0.54	-0.37	0.03	-0.26
austin	scarce	100.00	-1.16	-0.18	-0.08	0.01	-0.17
chicago	generous	93.80	-26.96	-14.75	-11.99	0.06	-2.02
chicago	moderate	100.00	-7.20	-3.83	-3.08	0.02	-0.61
chicago	scarce	100.00	-2.53	-1.02	-0.68	0.00	-0.12
nyc	generous	99.60	-58.30	-24.12	-22.03	0.01	-6.47
nyc	moderate	100.00	-13.00	-3.73	-3.48	-0.03	-1.37
nyc	scarce	100.00	-5.76	-1.26	-1.09	0.00	-0.63
sf	generous	92.10	-6.15	-6.36	-6.14	0.79	0.80
sf	moderate	99.80	-4.67	-4.47	-3.68	0.07	-0.91
sf	scarce	99.80	-2.59	-1.66	-1.23	0.02	-0.52

point forecasts can make the downstream allocation objective non-identifying, and a genuine predictive distribution is one of two remedies for that non-identification — the other being an explicit secondary rule — though the two address the same mechanism and need not produce identical simulated loss. This is not an artifact of the model's miscalibration: split-conformal recalibration lifts the quantile model's 90% interval coverage from 77–81% to 88–90% (Section 6) yet changes the full-arm simulated loss by a median of -0.1% (at most 0.9%), so the effect reflects identification of the objective, not the particular dispersion of the intervals.

c) Forecast quality is associated with lower simulated decision loss within a fixed policy.: Replacing the trailing-mean baseline with the validation-selected forecaster, policy held fixed, lowers the cumulative unresolved stock in all 12 settings (average daily reduction in unresolved *stock* from ≈1.8 thousand request-equivalents in Austin to ≈72 thousand in New York under scarcity). Consistently, model *rankings* by test MAE and by simulated decision loss agree closely across the eleven-configuration grid (Spearman $\rho = 0.88$–0.99 in every city-regime cell): in this environment, accuracy is largely a sufficient proxy for simulated decision value among reasonable models.

d) Accuracy-based and decision-based selection mostly agree.: Selecting over the cross-scope eleven-configuration grid by validation decision loss instead of validation MAE — both under the frozen budgets — yields the same choice in 9 of 12 settings, a gain in 2 (Austin, scarce and moderate: validation MAE prefers the pooled model there, the decision criterion prefers the local one, and the local choice is better on the held-out test), and one interval-tie; it did not increase test loss in any cell of this pre-specified grid. In San Francisco both criteria select the pooled model, which is also the better test model there. Decision-based validation selection is therefore a low-cost diagnostic rather than an established improvement.

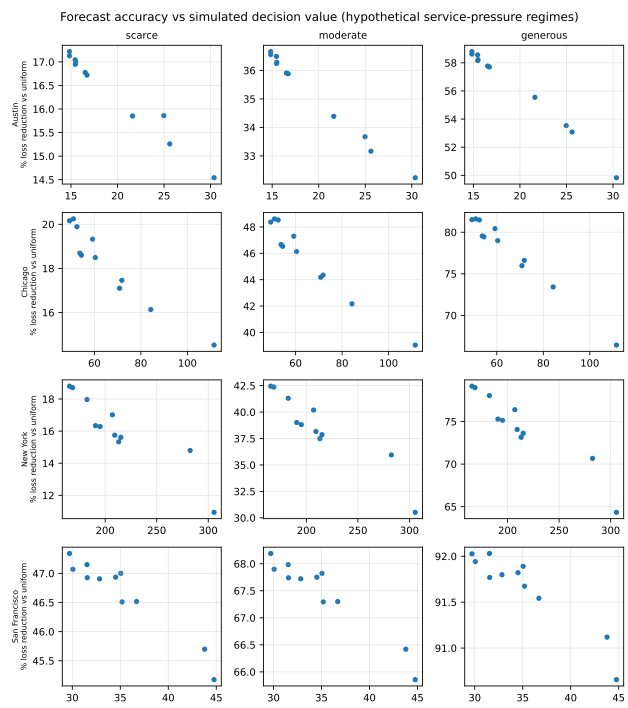

Fig. 4. Forecast accuracy (test MAE, x) versus simulated decision value (% loss reduction relative to the uniform floor, y) for the pre-specified point-forecast grid under the greedy expected-value policy with the fixed-index tie-break — the non-identified point-forecast policy of this section, shown as a diagnostic — by city and hypothetical service-pressure regime.

VIII. ROBUSTNESS AND SENSITIVITY

a) Trend diagnostic and inference framing.: For every city, regime, and pre-named contrast we regress the daily loss differential on the day index and test the slope with a Newey–West (lag-28) statistic, reporting also the share of the mean differential attributable to the linear drift (trend-diagnostic table, supplement). The result is unambiguous: 45 of 48 differential series are trend-dominated (the only stationarity- compatible cell is Chicago under generous capacity), and 11 of 12 loss *level* series trend as well. This is a structural property of the carryover stock, strongest under scarcity but present broadly: policy differences compound rather than fluctuate around a stationary mean. Accordingly, all decision-contrast intervals in Section 7 are read as descriptive characterizations of a systematic divergence within the deterministic simulation; a stationary-inference reading is retained only for the Chicago/generous cell. The forecast-accuracy contrasts of Section 6 are unaffected — their daily error differentials carry no stock — and their inferential reading stands.

b) Block-length stability.: None of the 48 decision-contrast interval conclusions changes when the bootstrap block length moves from the primary 28 days to 14 or 56 days; the same holds for the forecast contrasts. (Block-length stability does not, by itself, detect a common trend — which is why the explicit trend diagnostic above is the binding check.)

c) Decision-layer sensitivity.: The pre-specified grid preserves every qualitative conclusion: with 10% daily abandonment, policy ordering is unchanged in all cities; with one-family-at-a-time service-yield multipliers of 0.70 and 1.30 over every active family, the ordering direction never reverses (the largest perturbation is New York, generous regime, noise yield $\times 1.30$); with budgets calibrated on train+validation instead of training only, conclusions are unchanged; and the Austin analysis with the heterogeneous *other* family excluded (budgets recomputed) produces the identical policy ordering as the primary analysis, so Austin's conclusions do not depend on its residual category. Initial carryover: replacing the primary $b_0 = 0$ with a validation warm-up carryover or with each family's training-average daily demand leaves the claim-bearing ordering (uniform worst; the fixed-index point-greedy below the three identified policies) unchanged in all 12 city-regime cells (initial-carryover table, supplement). Under the illustrative normative-priority scenario the proportional–optimizer gap compresses, which is why the proportional-dominance claim is stated as conditional on the equal-weight objective.

d) What would change the conclusions.: The simulation results are conditional on the stylization assumptions (interchangeable units, fixed yield, no intra-day dynamics) and on the equal-weights objective; we claim regime *patterns* within this simulated environment, not city-specific magnitudes. The forecasting results are unconditional on those assumptions; the weather findings are conditional on day-ahead forecast availability (A3), bounded by the lagged-weather variant.

IX. Responsible Use of Reported-Demand Data

311 records measure the propensity to report as much as the incidence of problems: communities differ systematically in access, trust, language, and expectation that reporting leads to action [1, 8], and prediction-driven prioritization can institutionalize whatever bias the data carry [25]. Three design consequences follow in this study. First, the target variable is named throughout as *reported* demand, and no result is interpreted as a measurement of need. Second, because our spatial unit is the whole city, the most direct channel for reporting-bias amplification — reallocating resources *away* from low-reporting neighbourhoods — is outside the action space of the simulated planner; the family-level channel remains, and we report per-family served fractions and unserved-stock shares for every policy (supplement) so that systematic deprioritization of any family is visible rather than hidden in aggregate scores. These tradeoffs are real: under the equal request-equivalent objective the forecast-driven full-distribution policy attains low aggregate loss while serving a near-zero fraction of the low-volume water/sewer family in all four cities (and of public safety in San Francisco), because minimizing total request-equivalent stock concentrates capacity on high-volume families. Lower aggregate simulated loss is therefore not a fairness guarantee. Third, the priority weights ω_s encode a value judgment, not a fact; we expose them as configuration, vary them in sensitivity analysis, and regard their selection in any real deployment as a decision that belongs to accountable public officials, not to a model.

X. Limitations

- **Explicit simulation boundary.** The allocation layer is a transparent simulation over abstract request-equivalent capacity units; it omits shifts, travel, specialization, geography, and intra-day dynamics, and none of its parameters estimates any city's actual capacity, productivity, or queues. Conclusions concern the structure of the accuracy-to-allocation mapping inside this simulated environment.
- **Queue instability and horizon dependence.** Under the scarce (and to a lesser degree moderate) regimes the simulated system is structurally under-capacitated, so the unresolved stock grows over the evaluation horizon; cumulative-loss magnitudes and percentage reductions therefore depend on the horizon length, regime comparisons are partly mechanical (differences compress under supercriticality), and decision-contrast intervals are descriptive rather than stationary inference (Section 8).
- **Objective dependence.** The proportional-dominance result is conditional on the equal request-equivalent-weights objective; the normative-weight sensitivity compresses the gap. Equal weights are a deliberately simple choice, not a normatively neutral one, and no weighting here is socially validated.
- **Reported demand only.** Targets and losses are functions of reported requests; unreported need is invisible to both the forecasters and the simulated planner (Section 9).
- **Harmonization is a modeling choice.** The service families are deterministic keyword buckets; native taxonomies differ across cities (Chicago lacks a residential-noise intake type; Austin's residual *other* is large, though its exclusion leaves conclusions unchanged). All mappings and compositions are published.
- **Weather information set.** Target-day weather uses realized observations as a proxy for a day-ahead forecast (A3), bounded by the lagged-weather variant; one station per city coarsens spatial variation (A2); wind was excluded for data-completeness reasons.

- **Calibration.** The quantile model's intervals are under-dispersed (90% nominal, 77–81% empirical); a split-conformal recalibration (restoring 88–90% coverage) leaves the simulated-allocation result essentially unchanged (Sections 6–7), so the decision finding does not rest on calibrated uncertainty but on identification of the objective.
- **Scope.** Four U.S. Socrata-based systems; fixed a priori hyperparameters; no deep or foundation models; no causal claims; no deployment; no measurement of social need.

XI. Conclusion

We built a guarded, reproducible four-city benchmark of next-day reported 311 demand and coupled it to a controlled, explicitly simulated capacity-allocation evaluation with frozen pre-test environments and validation-only selection; the benchmark and its protocol are the primary contribution. The headline finding is what the coupled simulation reveals: a degenerate point forecast leaves the equal request-equivalent allocation objective non-identified, so its realized outcome is determined by the secondary tie-break rule (a proportional secondary rule recovers proportional performance and the quantile-interpolated predictive distribution recovers most of the fixed-index gap, under the tested regimes and objective). The supporting forecasting evidence is precise: public calendar signals improve next-day forecasts in every system and weather helps heterogeneously; temporally valid pooling and transfer mostly hurt; and decision-aware validation selection mostly coincides with accuracy-based selection, without increasing test loss in the pre-specified grid. These aggregate simulated-loss reductions can nonetheless hide family-level tradeoffs: some low-volume families are served at near-zero fractions under this objective. The broader lesson for forecast-driven public-sector analytics is methodological: evaluation environments must be frozen before selection, selection must never touch test outcomes, the interface between a forecast and a policy deserves the same scrutiny as the forecast itself, and reproducible benchmarks should report boundaries and failure modes, not only aggregate gains. Natural continuations include richer service-time modeling, spatial disaggregation, and extension beyond U.S. systems.

a) Reproducibility statement.: All data are public; the companion repository contains acquisition scripts with exact query URLs, SHA-256 manifests for every raw file, deterministic seeds, unit tests (including leakage and allocation-optimality checks), and a `Makefile` that regenerates every number, table, and figure in this paper from the versioned raw data with a single command.

References

[1] C. E. Kontokosta and B. Hong, "Bias in smart city governance: How socio-spatial disparities in 311 complaint behavior impact the fairness of data-driven decisions," *Sustainable Cities and Society*, vol. 64, p. 102503, 2021.

[2] A. N. Elmachtoub and P. Grigas, "Smart "predict, then optimize"," *Management Science*, vol. 68, no. 1, pp. 9–26, 2022.

[3] J. Mandi *et al.*, "Decision-focused learning: Foundations, state of the art, benchmark and future opportunities," *Journal of Artificial Intelligence Research*, vol. 80, pp. 1623–1701, 2024.

[4] R. Cui, S. Gallino, A. Moreno, and D. J. Zhang, "The operational value of social media information," *Production and Operations Management*, vol. 27, no. 10, pp. 1749–1769, 2018.

[5] S. Steinker, K. Hoberg, and U. W. Thonemann, "The value of weather information for e-commerce operations," *Production and Operations Management*, vol. 26, no. 10, pp. 1854–1874, 2017.

[6] L. Xu, M.-P. Kwan, S. McLafferty, and S. Wang, "Predicting demand for 311 non-emergency municipal services: An adaptive space-time kernel approach," *Applied Geography*, vol. 89, pp. 133–141, 2017.

[7] S. Cheng, S. Ganapati, G. Narasimhan, and F. B. Yusuf, "A machine learning-based analysis of 311 requests in the Miami-Dade county," *Growth and Change*, vol. 53, no. 4, pp. 1627–1645, 2022.

[8] C. E. Kontokosta, B. Hong, and K. Korsberg, "Equity in 311 reporting: Understanding socio-spatial differentials in the propensity to complain," *arXiv preprint arXiv:1710.02452*, 2017.

[9] D. Bertsimas and N. Kallus, "From predictive to prescriptive analytics," *Management Science*, vol. 66, no. 3, pp. 1025–1044, 2020.

[10] P. L. Donti, B. Amos, and J. Z. Kolter, "Task-based end-to-end model learning in stochastic optimization," in *Advances in Neural Information Processing Systems 30*, 2017, pp. 5484–5494.

[11] B. Tang and E. B. Khalil, "PyEPO: A PyTorch-based end-to-end predict-then-optimize library for linear and integer programming," *Mathematical Programming Computation*, vol. 16, pp. 297–335, 2024.

[12] U. Sadana, A. Chenreddy, E. Delage, A. Forel, E. Frejinger, and T. Vidal, "A survey of contextual optimization methods for decision-making under uncertainty," *European Journal of Operational Research*, vol. 320, no. 2, pp. 271–289, 2025.

[13] L. V. Green, P. J. Kolesar, and W. Whitt, "Coping with time-varying demand when setting staffing requirements for a service system," *Production and Operations Management*, vol. 16, no. 1, pp. 13–39, 2007.

[14] B. Rostami-Tabar and R. J. Hyndman, "Hierarchical time series forecasting in emergency medical services," *Journal of Service Research*, vol. 28, no. 2, pp. 278–295, 2025.

[15] Y. Zheng, L. Capra, O. Wolfson, and H. Yang, "Urban computing: Concepts, methodologies, and applications," *ACM Transactions on Intelligent Systems and Technology*, vol. 5, no. 3, pp. 38:1–38:55, 2014.

[16] L. Wang, X. Geng, X. Ma, F. Liu, and Q. Yang, "Cross-city transfer learning for deep spatio-temporal prediction,"

arXiv preprint arXiv:1802.00386, 2019, also presented at IJCAI 2019.
[17] D. Salinas, V. Flunkert, J. Gasthaus, and T. Januschowski, "DeepAR: Probabilistic forecasting with autoregressive recurrent networks," *International Journal of Forecasting*, vol. 36, no. 3, pp. 1181–1191, 2020.
[18] X. Liu *et al.*, "LargeST: A benchmark dataset for large-scale traffic forecasting," in *Advances in Neural Information Processing Systems 36 (NeurIPS 2023), Datasets and Benchmarks Track*, 2023.
[19] Y. Ning, H. Liu, H. Wang, Z. Zeng, and H. Xiong, "UUKG: Unified urban knowledge graph dataset for urban spatiotemporal prediction," in *Advances in Neural Information Processing Systems 36 (NeurIPS 2023), Datasets and Benchmarks Track*, 2023.
[20] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
[21] G. Ke *et al.*, "LightGBM: A highly efficient gradient boosting decision tree," in *Advances in Neural Information Processing Systems 30*, 2017, pp. 3146–3154.
[22] R. Koenker and G. Bassett, "Regression quantiles," *Econometrica*, vol. 46, no. 1, pp. 33–50, 1978.
[23] R. J. Hyndman and G. Athanasopoulos, *Forecasting: Principles and Practice*, 3rd ed. Melbourne: OTexts, 2021.
[24] F. X. Diebold and R. S. Mariano, "Comparing predictive accuracy," *Journal of Business & Economic Statistics*, vol. 13, no. 3, pp. 253–263, 1995.
[25] M. Samorani, S. L. Harris, L. G. Blount, H. Lu, and M. A. Santoro, "Overbooked and overlooked: Machine learning and racial bias in medical appointment scheduling," *Manufacturing & Service Operations Management*, vol. 24, no. 6, pp. 2825–2842, 2022.